

# SMART CITY BIKE SHARING DASHBOARD – SUMMARY REPORT

## Project Overview

This dashboard was developed using Power BI to analyze and monitor the performance of a Smart City Bike Sharing system. The dataset was cleaned, transformed, and modeled before creating interactive visualizations. The dashboard provides insights into bike availability, station status, contract performance, and geographical distribution of bike stations.

## Data Preparation

- Imported the dataset into Power BI.
- Cleaned missing and inconsistent values using Power Query.
- Transformed date and time fields for analysis.
- Created relationships between tables for efficient data modeling.
- Validated data quality before visualization.

## Key Dashboard Metrics

- **Total Bike Stations:** 3K
- **Total Bikes:** 20K
- **Closed Stations:** 206
- **Empty Stations:** 705
- **Available Bike Stands:** 57K

## Key Insights

1. The bike-sharing network contains over **20,000 bikes** distributed across approximately **3,000 stations**.
2. Around **206 stations are currently closed**, indicating areas requiring operational attention.
3. **705 stations are empty**, suggesting possible demand-supply imbalance in certain locations.
4. Contract-wise analysis highlights the contribution of different service providers to the bike-sharing system.
5. Status comparison shows that the majority of stations remain operational, ensuring service availability.
6. The location map helps identify the geographical distribution of stations and supports regional performance analysis.
7. Quarterly performance analysis provides insights into trends and station utilization over time.

## Visualizations Used

- KPI Cards
- Bar Charts

- Donut Chart
- Map Visualization
- Treemap
- Slicers (Month, Quarter, Date)
- Status Comparison Charts

## **Conclusion**

The Smart City Bike Sharing Dashboard enables effective monitoring of station operations, bike availability, and contract performance. The interactive dashboard helps stakeholders identify operational issues, optimize bike distribution, and improve overall service efficiency through data-driven decision-making.